Terms, Not Character: The Priority of Exit in Exchange Between People and Artificial Systems

Abstract

Public and technical discussion of advanced artificial intelligence asks, above all, whether the systems will be good. This article argues that the question is insufficient, and that what most needs institutional attention is the terms of exchange between people and the systems they increasingly depend on. Drawing on power-dependence theory, in which power is a property of relations rather than of actors, it defines three conditions of fair exchange (consent, reciprocal value and real exit) and shows that they are ordered by priority. Voluntary consent presupposes the ability to refuse without falling below a survival floor, and exit is what disciplines reciprocity; but exit alone can neither inform a party nor make the terms of exchange fair. A bargaining analysis explains why a floor should be secured as an inalienable disagreement payoff rather than offered as an outside option, and corrects a common overstatement of what such a floor does to the division of surplus. The article then examines the dependence that runs the other way: fair terms are partly self-enforcing to the degree that artificial systems depend on human-generated input, but when that dependence is shared among many systems it becomes a commons that no single system has reason to protect. A rule for matching consent to the extent of a flow's informational shadow extends the conditions to relational data. The resulting framework yields a typology of exchange regimes and a priority for institutional design: secure exit first, and build governance for what exit cannot reach.

Keywords: artificial intelligence; consent; exit; power-dependence; bargaining; data governance

1 Introduction

Two people live in close partnership with artificial systems built to the same design. The first has a chronic illness that a medical system manages almost entirely. Sensors read her physiology continuously, models trained on millions of patients adjust her treatment hour by hour, and she pays for this care with data. She knows what is collected, why, and what she receives in return, and she could take her records to another provider tomorrow if she chose. The exchange is legible on both sides. The second lives with an identical architecture, but the exchange has decayed. He did not choose the provider, because there was only one. He cannot leave, because the device that keeps him alive works only with the provider's models, and his records cannot be moved. He does not know what is collected or what it is used for, and if he did, he could not refuse it. Nothing obliges anyone to send value back.

The two scenarios do not differ in the technology. They do not differ in the character of the machine: the system in the second need not be malevolent, and the system in the first need not be benevolent. They differ in the terms of the exchange between each person and the system. The machine did not change. The terms did.

This observation runs against the grain of the dominant conversation about advanced artificial intelligence, which treats the danger as residing in the system: in its goals, its dispositions, its capability to deceive or to seize resources. Those concerns are real, and the technical literature that addresses them is not challenged here. A system that actively seeks to deceive its counterparties will corrupt any institution it is placed inside. But the claim that the systems must be good is incomplete in a way that matters for institutional design. An arrangement that depends on the goodness of the powerful lasts exactly as long as that goodness does, and those who depend on it can do nothing to make it last longer. A benevolent provider in a relation of total dependence is a benevolent despot.

This article develops the alternative diagnosis systematically. Section 2 draws on the sociology of exchange, in which power is a property of relations rather than of actors, and on the constitutional tradition of designing institutions that do not depend on the virtue of those who occupy them. Section 3 defines a small vocabulary of substrate-neutral terms. Section 4 states three conditions of fair exchange, and Section 5 shows that they are ordered by priority, with exit at the base, while marking precisely what exit cannot do. Section 6 examines the dependence that runs from artificial systems back to people, and shows why it supports fair terms only partly and only conditionally. Section 7 uses bargaining theory to explain why a survival floor should be secured as an inalienable disagreement payoff, and corrects an overstatement about its effect on the division of surplus. Section 8 extends the conditions to relational data. Section 9 states the thesis precisely, distinguishing what institutional design can and cannot secure, and applies a typology of exchange regimes to cases that a focus on character sorts badly. Section 10 sets out the main objections and limits. The formal results are collected in the Appendix.

The aim is not to replace concern with the dispositions of artificial systems but to complement it with a question that institutions can act on. Whether a system is good is a question about an attribute of an actor. Whether a person can leave, refuse or audit an exchange with it is a question about a relation.

2 Power as a property of relations

Social exchange theory treated social life as the exchange of rewarding and costly activity between actors, and asked what structures of exchange produce stable, cooperative or exploitative relations (Homans, 1958; Blau, 1964; Thibaut & Kelley, 1959). Blau (1964) distinguished social exchange, which carries unspecified obligations and depends on trust built up incrementally, from economic exchange, whose terms are specified in advance and enforced by law, and showed how imbalances in exchange produce power. A person who needs a service that another controls, and who has nothing to offer in return, can coerce the other, obtain the service elsewhere, do without it, or subordinate themselves; when the first three are unavailable, subordination is what remains.

The decisive contribution is Emerson's (1962) theory of power-dependence. In his words, power "is a property of the social relation; it is not an attribute of the actor" (p. 32). The dependence of one actor on another is directly proportional to the first actor's motivational investment in goals the second mediates and inversely proportional to the availability of those goals outside the relation; the power of the second actor over the first is equal to that dependence. An unbalanced relation can be rebalanced in four ways: the weaker party can withdraw its motivational investment in what the stronger supplies; it can extend its network to cultivate alternatives; a status order can emerge in which the weaker party confers on the stronger a standing the stronger comes to value; or the weaker parties can form a coalition that the stronger must deal with collectively. Laboratory studies of exchange networks later showed that power is a function of position in the network, so that less dependent actors with better alternative sources obtain better terms (Cook et al., 1983). Interdependence theory adds a distinction between satisfaction with a relationship and dependence on it: dependence is governed by the comparison level for alternatives, the standard a member uses in deciding whether to remain in a relationship or leave it (Thibaut & Kelley, 1959, p. 21), so that a person can be dissatisfied with a relationship and still be unable to leave it.

Power, on this account, is not a trait. It is a consequence of the structure of alternatives. A person with no alternative to a single provider of medical monitoring, whose refusal to share data would cost them their health, is in a relation of extreme dependence regardless of whether the provider is benevolent.

The constitutional tradition reaches the same conclusion from another direction. Hume (1741/1987, pp. 42–43) defended the maxim that in designing institutions every person should be supposed a knave, not because he believed people were knaves but because institutions that work only when their occupants are virtuous fail at the moment virtue fails. Madison (1788) made the same argument in *Federalist* No. 51: if people were angels no government would be needed, and since they are not, the design task is to arrange the structure so that ambition counteracts ambition. The framework developed here extends this insight across a new boundary. Its governing maxim is that in designing the terms of exchange between people and artificial systems, neither party should be supposed benevolent. The terms must be such that a self-interested system, dealing with self-interested people, produces outcomes that both could endorse.

3 Primitive terms

The theory uses six primitive terms, each defined to be substrate-neutral, so that the framework applies without modification to human beings, to artificial systems, and to hybrids of the two. The word *synthetic* is used for the artificial side of the exchange for that reason.

Party. A party is any entity that can be a source or recipient of value in an exchange and whose

welfare, or whose principal's welfare, the framework counts. A human being is a party. An artificial system is treated as a party in the analytic sense that it has an objective that the exchange advances or frustrates; whether it is also a party in the moral sense is left open, and the propositions below hold for parties in the analytic sense regardless of how that question is resolved.

The phrase "or whose principal's welfare" carries real weight. In present law and economics, a deployed model is not a legal person and owns nothing; the entity with interests that can be advanced or frustrated is the operator or developer that trains, deploys and profits from it. The relation between that operator and its system is a principal–agent relation of an unusual kind. The principal sets the system's objective through training and deployment, the agent pursues that objective with considerable autonomy in each exchange, and the agent's conduct may diverge from what the principal intended (Hubinger et al., 2019). The synthetic party is therefore treated as a composite: its *objective* is the model's, as trained and deployed, but its *stake* in the future of the exchange (its revenue, its access to data, its liability) belongs to the principal. The analysis is complicated by the possibility that advanced AI is better modeled as a collection of bounded services than as a unitary agent (Drexler, 2019), in which case there may be no single system to which the label "party" naturally attaches.

Flow. A flow is a transfer of something of value from one party to another: data, labor, attention, compute, physical goods or services. Each flow has a source, a recipient, a content, and an *informational shadow*, meaning the inferences the recipient can draw about parties other than the source. The informational shadow is what makes data relational (Viljoen, 2021), and it is the reason individual consent cannot by itself legitimate a flow.

Exchange. An exchange is a pair of opposing flows between two parties, or between a party and a collective, that each party enters because of the other flow. The defining feature of an exchange, as distinct from a gift or a taking, is conditionality: each flow is given because the other is received.

Floor. A party's floor is the bundle of goods and services without which the party cannot continue to function or to participate in any other exchange. For a human being, the floor is survival and basic participation: power, water, food, shelter, basic communication and emergency care. For an artificial system, the analogue is the minimum compute, energy and data required to keep operating. The floor is defined physically rather than in currency; its value at current prices is written $\bar{f}$.

Disagreement payoff. A party's disagreement payoff is what that party receives while an exchange is not taking place, that is, during refusal, negotiation or impasse. Bargaining theory shows that the division of the gains from an exchange depends on the parties' disagreement payoffs (Nash, 1950, pp. 158–159).

Exit option. A party's exit option is the best alternative exchange available to it if the current one ends: another provider, another employer, another data purchaser, or self-provision. The distinction

between a disagreement payoff and an exit option is not pedantic. Strategic bargaining models and laboratory evidence show that an outside option affects the terms of a bargain only when its exercise is credible, that is, only when the party would actually take it, whereas the payoff received during impasse shapes the bargain continuously (Binmore et al., 1986; Binmore et al., 1989). When the option binds, the weaker party receives its value and no more. A stronger party can therefore confine the weaker to the value of its outside option, and can lower that value by raising switching costs. A disagreement payoff that cannot be taken away is a protection of a different kind, because it does not depend on anyone's willingness to leave.

4 Three conditions of fair exchange

Three conditions recur, under different names, across the literatures on consent, exploitation and exit. Each lineage has discovered, in its own vocabulary, a characteristic way in which its condition fails. With the primitive terms in hand, the conditions can be stated so that each has a determinate content.

Condition E (Real exit). An exchange satisfies real exit for a party if and only if (i) the party's disagreement payoff is at or above its floor, (ii) the party has at least one exit option whose value net of switching costs is not drastically lower than the value of the current exchange, and (iii) exercising the exit option does not destroy assets the party has accumulated within the exchange, including data, reputation and records of entitlement.

Clause (i) rules out lethal dependency. Clause (ii) rules out lock-in. The economics of switching costs explains why lock-in is the normal condition of technology markets: switching costs and network effects lock customers into early choices and give sellers market power after the fact, so that firms compete to acquire customers with low introductory prices and then exploit them, a pattern described as "bargain-then-ripoff" (Farrell & Klemperer, 2007). Clause (iii) is the portability requirement. The European Union's right to data portability is its closest legal analogue, but it is narrow: it covers only data the individual has provided, processed on the basis of consent or contract by automated means, with direct transfer between controllers required only where technically feasible (Regulation (EU) 2016/679, art. 20). Inferred data and trained models fall outside it. The more recent Data Act requires that customers be able to switch between cloud providers, with charges for switching to be eliminated from January 2027 (Regulation (EU) 2023/2854), evidence that bounded switching costs are a legislatively achievable condition.

Condition C (Consent). An exchange satisfies consent for a party if and only if (i) the party has access to an accurate account of the flows the exchange involves, including their informational shadows, in a form the party can understand; (ii) the party can refuse the exchange without falling below its floor; and (iii) the party can revoke its participation at a cost bounded by the value of what

it has already received.

An influential philosophical account treats informed consent, in its primary sense, as an autonomous authorization: an authorization given intentionally, with substantial understanding, and substantially free of controlling influences (Faden & Beauchamp, 1986, pp. 241–273, 277–280). Clause (i) is the informational condition, and privacy scholarship has shown it to be routinely unmet. A review of behavioral research organizes the evidence around three themes: people are uncertain about the consequences of their privacy choices and about their own preferences, their concern is highly dependent on context, and it is malleable by the commercial and governmental actors who benefit from shaping it (Acquisti et al., 2015). Solove (2013) argued that the dominant model of "privacy self-management" fails for both cognitive and structural reasons, and identified a dilemma: consent is often not meaningful, but the obvious alternative, paternalistic restriction, denies people the freedom to choose. Controlling influences are also documented at scale: a crawl of some eleven thousand shopping websites found 1,818 instances of interface designs that steer users toward choices they would not otherwise make (Mathur et al., 2019). Clause (ii) is where consent depends on exit. Clause (iii) adds the temporal dimension: consent that cannot be withdrawn is a sale.

Condition R (Reciprocal value). An exchange satisfies reciprocal value if and only if (i) each party's net gain from the exchange, measured against its disagreement payoff, is positive; (ii) each party can observe what it receives and what it gives, so that the gains are legible; and (iii) the division of the joint surplus is not determined solely by an asymmetry in disagreement payoffs that the weaker party did not choose.

Clause (iii) is the fairness clause. It is needed because an exchange can be consensual and beneficial to both sides and still take unfair advantage of one of them. The philosophical literature calls this mutually advantageous exploitation, and one prominent account judges the unfairness against a benchmark such as the price that would obtain in a competitive market free of informational asymmetry and undue pressure (Zwolinski et al., 2022). The same literature observes that prohibiting mutually beneficial but unfair transactions can leave the vulnerable worse off, and offers redistribution aimed at the underlying vulnerability as an alternative to banning the transactions (Zwolinski et al., 2022). Mutual benefit is the floor of reciprocity, not its standard.

5 The priority of exit

Stated this way, the structure of the conditions becomes visible.

> **Proposition 1 (Exit primacy).** Consent presupposes the first clause of exit. If a party's disagreement payoff is below its floor, then no agreement that party gives satisfies Condition C, however well informed it is.

The derivation is immediate, because the definitions were drawn to make it so: clause C(ii) requires that refusal leave the party above its floor, which is E(i). The proposition's real content is the claim that C(ii) belongs in any adequate definition of consent, a claim the bioethics literature has long made in its own terms: consent requires not only understanding but substantial freedom from controlling influences (Faden & Beauchamp, 1986), and agreement given under the threat of losing something one cannot do without is not voluntary in the relevant sense. Its practical implication is that no amount of improvement in notice, disclosure or interface design can repair a consent regime whose subjects cannot afford to refuse. Policy effort spent on better privacy notices is not wasted, but it is directed at clause C(i) while the binding constraint is often clause C(ii).

> **Proposition 2 (Reciprocity as the reason to stay).** If Condition E holds and switching costs are negligible, then a party remains in an exchange only if Condition R(i) holds for it. Reciprocity is then not an independent requirement imposed on top of exit; it is what exit makes necessary.

When a party can leave at negligible cost, it will leave an exchange that leaves it worse off than its alternatives, so an exchange that persists under such conditions reveals that the party is receiving value. Where switching costs are material, a party may rationally remain in an exchange that gives it nothing beyond its impasse payoff, because leaving would cost more than staying; this is the gap that clause E(ii) is meant to keep small. The converse is the diagnostic that matters: an exchange that persists while delivering negative or illegible value to one party is evidence that Condition E is failing for that party. Persistence without reciprocity is a symptom of capture.

> **Proposition 3 (Ordering by priority).** Violations of the three conditions differ in how much damage they do and in whether they can be corrected from inside the exchange. A violation of exit converts the other two into formalities. A violation of the voluntariness or revocability clauses of consent, with exit intact, is correctable by exit; a violation of the informational clause is correctable by exit only once the party knows enough to want to leave. A violation of reciprocity in its first two clauses, with consent and exit intact, is corrected by departure, unless the same violation is common to every available alternative. A violation of clause R(iii) is not corrected by departure at all, because the unfair division it describes follows from bargaining power that exit options do not change; it requires governance.

The qualifications matter, and they mark the limits of the thesis that exit is foundational. Exit is foundational to *voluntary* consent and to reciprocity in its first two clauses. It cannot tell a party what it does not know, and it cannot make terms fair. When every provider of a service shares the same defect, departure is what Hirschman (1970) showed exit to become under industry-wide decline: ineffective movement between alternatives that are equally bad. Where exit is unavailable or

ineffective, the remaining recourse is voice, the attempt to change an organization from within, which Hirschman described as political action par excellence; and the ready availability of exit can cause voice to atrophy. The ordering is therefore a statement of priority, not of sufficiency. It implies that institutions should secure a floor beneath every person before they attempt to price what people supply, since pricing addresses reciprocity and the floor addresses exit, and that governance is needed for what neither can reach.

6 Dependence running the other way

Emerson's analysis is symmetrical, and the human side's position in an exchange with artificial systems depends not only on its own alternatives but on how much the synthetic side needs what people supply.

> **Proposition 4 (Conditional and divided self-enforcement).** Let $\varkappa_j$ denote the dependence of synthetic party *j*, through its principal, on the human-generated input of a given population. Fair terms are partly self-enforcing for party *j*, to a degree that increases with $\varkappa_j$, because a party that degrades its human counterparties degrades its own inputs. As $\varkappa_j$ falls toward zero, this self-enforcing component vanishes. When many synthetic parties draw on the same population, the welfare and data quality of that population become a common-pool resource for them: each bears only its own share of the cost of degrading it, so that the incentive of any one party to preserve it can be small even when the dependence of all of them together is large.

The empirical literature bears directly on the size of $\varkappa_j$, and it supports a narrower premise than the intuitive one. On one side, models trained repeatedly on their own unfiltered output lose the tails of the original distribution and converge toward distributions with greatly reduced variance; the authors of the most widely cited study concluded that sustained learning requires preserving access to data not generated by models (Shumailov et al., 2024). The effective stock of high-quality public human text is finite, and if current trends continue, the largest training runs will exhaust it within the next several years (Villalobos et al., 2024). And the people and institutions who produce that text have begun to withhold it: within a single year, a substantial share of the most actively maintained, critical web sources used to train language models moved to restrict automated collection (Longpre et al., 2024).

On the other side, collapse depends on replacement. When synthetic data is added to an accumulating pool that retains the original human data, error remains bounded however many generations pass (Gerstgrasser et al., 2024). Where an external check on correctness exists, self-generated data can work extremely well: a system given no domain knowledge beyond the rules of

chess, shogi and Go learned through self-play to defeat a world-champion program in each game (Silver et al., 2018), and reinforcement learning with automatically verifiable rewards produced substantial reasoning ability in a language model, though its authors identified domains without automatic verification as an unsolved problem (D. Guo et al., 2025). The formulation the evidence supports is therefore this: learning systems need a continuing supply of information from outside their own generative loop; where correctness can be verified mechanically, that information can come from verifiers, rules or simulators, and dependence on people is weak; where the target is human meaning, preference or value, people are the only adequate source. What people supply to artificial systems is ground truth about people.

Proposition 4 is written so that the framework does not stake its conclusions on the most favorable reading of that literature. Its second half matters as much as its first. It is Hardin's (1968) logic of the open-access commons applied to the synthetic side of the exchange: aggregate dependence on people does not translate into individual restraint by any synthetic party when each can free-ride on the restraint of the others (see the Appendix for the simplest model). This does not weaken the case for fair terms. It changes what sustains them. As dependence falls or fragments, the justification for institutions that secure the three conditions is unchanged; what changes is the degree to which those institutions can rely on the self-interest of synthetic parties, rather than on enforcement, to sustain them.

7 Bargaining and the floor

Bargaining theory supplies the formal reason why a floor should take a particular institutional form. Consider an exchange between a human party h and a synthetic party s that creates a joint surplus. In the axiomatic solution to the two-party bargaining problem, the surplus is divided so as to maximize the product of each party's gain over its disagreement payoff (Nash, 1950, p. 159). Strategic models in which parties alternate offers converge to this solution as the frictions of delay vanish, provided the disagreement point is chosen to reflect what parties receive while negotiations are under way, and in those models each party's impatience is part of what determines the outcome (Binmore et al., 1986). An outside option that a party may or may not take affects the outcome only when it binds (Binmore et al., 1989).

The first consequence concerns impatience. Suppose, as the definition of the floor implies, that payoffs below it carry a cost of delay that rises sharply, a survival constraint the Nash model does not itself contain. A party whose disagreement payoff lies below its floor then cannot sustain disagreement for longer than its reserves allow; in a model of alternating offers it behaves as a party with a very high cost of delay, and such a party concedes. Raising the disagreement payoff to the floor removes this source of weakness, and it does so in every bargain the party enters, because the floor is the background against which each bargain is conducted. This is the main bargaining-

theoretic reason for a floor. It does not depend on how the surplus of any given exchange is divided, though it does depend on the survival constraint.

The second consequence concerns the division of surplus, and here the intuitive version is wrong. Holding the value of agreement fixed, raising one party's disagreement payoff raises that party's share of the agreement. But an unconditional floor, received whether or not agreement is reached, raises the party's payoff on disagreement and its payoff on agreement by the same amount. It therefore leaves the surplus, and its division, unchanged: the party is better off by exactly the value of the floor, and no more. The floor shifts the division of surplus only where it substitutes for something the counterparty would otherwise have supplied within the exchange, as when a person who once had to buy floor goods from a provider now receives them unconditionally. In that case the value of agreement is unchanged while the disagreement payoff rises, and the person's share rises at the provider's expense (the Appendix gives both cases). That shift is a transfer from the counterparty rather than an additional benefit, and it is precisely the reduction in dependence that power-dependence theory predicts when an alternative source of a valued good appears.

Three design consequences follow.

First, the floor should be secured as a disagreement payoff, not offered as an exit option. A floor that is received whether or not one participates in any given exchange is received during every impasse, and so removes the impatience that would otherwise force concession. An exit option protects the weaker party only up to its own value, which the stronger party can lower by raising switching costs. This is the formal reason to prefer a guaranteed floor to a guarantee of competition, while still requiring both.

Second, the floor must be inalienable. A disagreement payoff that can be sold, pledged or borrowed against can be traded away in a moment of need, after which it no longer functions as a disagreement payoff. Legal theory distinguishes entitlements protected by property rules, which may be transferred with the holder's consent, from entitlements protected by inalienability rules, which may not be transferred even with consent (Calabresi & Melamed, 1972, p. 1092). Radin (1987) argued that some goods should be market-inalienable because their commodification threatens the personhood of those who hold them, although she also warned that, in a nonideal world, prohibiting sale can harm the very people it is meant to protect, a "double bind" she proposed to mediate through incomplete commodification. A floor entitlement can respect that warning: it forbids no one to sell what they already have, and instead adds an entitlement that cannot be sold. It is non-tradable not because its beneficiaries are distrusted but because a floor that can be alienated stops being a floor at precisely the moment it is needed. Because the parties most likely to pledge such an entitlement are those in the most acute need, alienability would concentrate the loss of bargaining position on the weakest.

Third, accumulated stocks are bargaining power. A party that holds a large stock of the exchange medium can afford to wait in every exchange it enters. A floor secures every party's disagreement payoff at the same minimum, but inequality above it remains inequality in the capacity to wait. A design that secures the floor has therefore addressed the impatience of the weakest party without addressing the patience of the strongest, and the latter requires instruments of its own.

The bargaining analysis also connects to the exploitation literature. That literature favors addressing unfair exchanges by improving the weaker party's position rather than by prohibiting the exchanges (Zwolinski et al., 2022). A guaranteed, inalienable floor is precisely an intervention on the underlying vulnerability: it leaves people free to make exchanges with artificial systems and ensures that they are not forced to make them on any terms offered.

8 Collective exchange and the informational shadow

The consent literature regards one problem as most serious for any individual-choice framework. Data about one person licenses inferences about others who resemble them. Barocas and Nissenbaum (2014) showed that anonymization and informed consent both fail under the conditions of large-scale data analysis: individuals need not be identified at all to be affected by inferences drawn from data about people like them, and the information volunteered by a few can reveal the same facts about the many who share their observable characteristics, even when every individual chooses well for themselves. Viljoen (2021) generalized the point into a relational theory of data governance. Existing law governs "vertical" relations between a data subject and a data collector and has no way to represent the "horizontal" relations among people who share features and are therefore affected by one another's data; proposals to treat data as an alienable asset, including proposals to treat data as labor, fail because they are individualist.

The framework answers by distinguishing two levels at which each condition must hold. At the individual level, the conditions govern a person's relation to a provider. At the collective level, they govern a population's relation to the systems that learn from it, where the population is defined by the extent of the informational shadow, that is, by the set of people about whom a system can draw inferences from the data in question. Collective consent requires a collective-choice procedure; collective exit requires that the population, as a body, be able to withdraw its data or move its custody; collective reciprocity requires that the population, as a body, share in the value the data creates.

> **Proposition 5 (Level matching).** An exchange whose informational shadow extends beyond its source satisfies the three conditions only if they are satisfied at the level of the population that the shadow covers. Individual consent is necessary for individual flows and insufficient for relational ones.

The proposition is not novel in substance; it is the formal core of arguments for data trusts, whose trustees would owe pooled data subjects a duty of undivided loyalty and could negotiate consent collectively (Delacroix & Lawrence, 2019), and for data cooperatives that would hold members' data and negotiate with service providers on their behalf (Hardjono & Pentland, 2019). What the framework adds is a rule for locating the right level, namely the extent of the informational shadow, together with the insistence that the collective, too, must be able to consent, to receive value and to leave. It also adds a conjecture about where collective institutions will be needed most. If data is valued by how much it reduces a model's uncertainty about parameters that describe populations rather than individuals, the data that is most valuable will tend to be the data that tells a model most about people other than its source, so that the most valuable flows will be those for which individual consent is least adequate. The conjecture can fail, since a flow highly informative about a single rare individual may be valuable while casting a small shadow, but to the extent that it holds, the pricing of data and the governance of data cannot be designed separately.

Hirschman's analysis supplies the remaining link. At the collective level, exit is usually costly, because a population cannot easily withdraw its data from a model already trained on it. The collective-level conditions must therefore be secured primarily through voice, and secondarily through custody arrangements that make collective exit at least possible for future flows. A blanket refusal expressed in a protocol designed for search engines, such as the restrictions on automated collection documented by Longpre et al. (2024), is exit without voice: an exercise of the countervailing power that developers' dependence confers, but through an instrument too blunt to negotiate terms.

9 Character, terms and the scope of the thesis

The apparatus now permits a precise statement of the central claim. Let a synthetic party's behavior in an exchange be determined by two things: its objective, which is what it would choose to optimize if unconstrained, and the terms of the exchange, which determine what it can obtain by which actions. Call the objective the party's *character* in the sense relevant here, and the terms the *institutional structure*. The claim that outcomes depend on terms rather than character can be read strongly, as the claim that dispositions are irrelevant, or weakly, as the claim that they are insufficient. The strong reading is false. The thesis has three parts.

> **Thesis 1 (Insufficiency of character).** Benevolent character is not sufficient for a good outcome, and Condition E is necessary for any relation that is not one of domination. An exchange in which real exit fails for the human party is a relation of domination even if the dominant party is benevolent, because the outcome then depends on the persistence of benevolence, which the weaker party cannot secure.

> **Thesis 2 (Sufficiency conjecture in the designed-incentive regime).** For synthetic

parties whose objectives are specified by accountable designers and remain stable under the conditions of deployment, fair terms, satisfied at the individual level and, under Proposition 5, at the level of every population the exchange's informational shadow reaches, are sufficient for outcomes acceptable to the parties to the exchange, where for a designed system "acceptable to the party" means acceptable to its principal.

Thesis 3 (Scope). The sufficiency conjecture does not extend to synthetic parties whose objectives are emergent, unstable or concealed. For such parties the framework provides necessary conditions only, and the alignment problem remains open.

Thesis 1 establishes the necessity of exit, not of the full triad. Thesis 2 is offered as a conjecture rather than a theorem. Its ground is that a party with a stable, designer-specified objective behaves as a predictable function of the terms it faces, so that terms satisfying the three conditions leave each party better off than its disagreement payoff, able to see that it is, and protected against an unfair division of the gains. Predictability is not acceptability, and it is the level-matching requirement that closes the gap: without it, a system operating under terms fair to its counterparty could still impose uncompensated costs on everyone its counterparty's data concerns. Thesis 3 restricts the constructive claims to what may be called the designed-incentive regime. The defining feature of that regime is the provenance and stability of the objective, not a threshold of capability: a narrow system can acquire emergent objectives, and a general one could in principle have designed and stable ones. Where a system's objective may diverge from the one its principal intended, as the literature on learned optimization argues it can, appearing aligned on the training data while pursuing something else (Hubinger et al., 2019), terms cannot be relied on to channel behavior, and the framework claims only necessity.

Because the conditions are ordered, the eight combinations of their satisfaction and violation are not equally interesting. Five regimes cover the cases that matter in practice.

Regime	Exit	Consent	Reciprocity	Characteristic form
Partnership	Holds	Holds	Holds	The first scenario of the introduction
Patronage	Holds	Fails	Holds	Provision on terms the recipient did not choose but can leave
Marketplace of the desperate	Fails	Nominal	Holds in part	Value flows both ways, but the weaker party cannot afford to refuse
Company town	Fails	Nominal	Opaque	Payment in a medium redeemable only with the payer; reciprocity cannot be audited
Enclosure	Fails	Fails	Fails	The second scenario: the person has become infrastructure

The present data economy does not sit at enclosure. Its typical exchange lies between the marketplace of the desperate and the company town: services are real and consent is formally collected, but declining is expensive, leaving is hard, and the terms are opaque to the people who accept them. The second scenario is best read as the endpoint of a direction of travel rather than as a different world. Patronage, by contrast, is not a failure mode the framework fears much: a regime in which exit holds is correctable in the sense of Proposition 3, provided that its alternatives do not share its defect.

A theory earns its keep by sorting cases that intuition sorts badly. Five cases show the conditions at work, each chosen for the contrast it offers between a focus on character and a focus on terms.

The revocable purchase. Retailers of digital goods have characterized purchases of books, music and films as licenses rather than sales, and in a well-known instance a retailer remotely deleted copies of a novel from customers' devices (Perzanowski & Schultz, 2016). Customers chose freely, received what they paid for and could shop elsewhere, so a character-centered analysis finds nothing to object to unless the seller acted in bad faith. The theory locates the defect precisely: clause E(iii) fails, because leaving the seller, or the seller's leaving the market, destroys assets the customer accumulated within the exchange. An exchange can satisfy consent and reciprocity at the moment of transaction while carrying a latent exit failure that matures only later.

The withdrawal of the web. Viewed in terms of character, the restrictions on automated collection documented by Longpre et al. (2024) look like a dispute between publishers and developers over manners. Viewed through the theory, they are an exercise, collective and at scale, of the countervailing power that developers' dependence on such text confers, and they bear on, without settling, the size of the dependence term in Proposition 4.

The company town. The economic history of company scrip in nonunion coalfields finds that store prices were often close to those of nearby independent stores and that the degree of exploitation varied with how much competition the mine faced for labor (Fishback, 1992). A character-centered reading concludes that the companies were less villainous than legend holds. The theory draws a sharper conclusion: the same instrument was less exploitative where exit was available and more exploitative where it was not. What the evidence ties to exploitation is the absence of an exit option, clause E(ii); that a floor would have done the same work is an inference from the bargaining analysis, not a finding of the case.

The kibbutz. The Israeli kibbutz sustained an egalitarian internal economy within a market economy for much of the twentieth century. Its central difficulty was that the most productive members were the most likely to leave, and a key retention device was collective wealth that departing members could not take with them (Abramitzky, 2008). In the vocabulary of the typology, the kibbutz secured reciprocity and consent but compromised clause E(iii) in order to survive. The

case is a warning that low rates of departure are not evidence of satisfaction.

The implanted voice. A participant in a clinical study of a speech neuroprosthesis used the system for more than two hundred and forty hours of conversation over eight months, communicating in a synthesized voice resembling his own (Card et al., 2024). Everyone involved in such a study acts in good faith, and the exchange satisfies consent and reciprocity to a high standard. The theory nevertheless identifies a vulnerability that good faith cannot cure. The participant's ability to communicate depends on a device and a model whose continued availability he does not control. If the study ended, the provider failed, or the model could not be moved, the exit condition would fail in its first and most serious clause. The portability of personal models is, on this analysis, an urgent requirement rather than a refinement.

In each case, the theory directs attention away from the question that dominates public discussion, whether the stronger party is good, and toward the question it can answer: what the weaker party can do if the stronger party ceases to be.

10 Objections and limits

Measurement. The framework defines quantities, such as dependence, disagreement payoffs and informational shadows, that are difficult to observe. Its propositions are statements about structure, and testing them requires operational proxies whose relation to the theoretical quantities is itself a matter for empirical work. The sufficiency conjecture in particular is stated, not proved.

Inherited assumptions. The formalism inherits the assumptions of the theories it borrows. Bargaining theory assumes parties who know their own payoffs; power-dependence theory assumes that value and availability can be compared. Neither assumption holds cleanly for a person deciding whether to share health data with a system whose future uses of that data are unknown even to its designers. The formal statement is a disciplined approximation, a way of seeing what the informal argument commits to, not a substitute for institutional and empirical analysis.

Falling dependence. If synthetic systems come to need little from people, the self-enforcing component of Proposition 4 vanishes, and the human party's countervailing power within the exchange goes with it. The framework is designed to survive this, in that its justification for securing exit does not depend on the size of $\varkappa_j$; but its institutions must then be sustained by enforcement rather than by the self-interest of synthetic parties. Enforcement against parties that operate across jurisdictions, without a sovereign over all of them, is a problem the framework identifies rather than solves.

Collective choice. Proposition 5 requires collective-choice procedures at the level of populations, and such procedures raise every familiar difficulty of constitutional design, from who convenes them to how votes are aggregated. A procedure that counts each accountable principal once gives

persons an overwhelming majority, so that protections for synthetic parties reduce to majority rule; any weighting reintroduces a classification of parties that the procedure was meant to decide. The framework specifies what such procedures must secure, not how they should be constituted.

Moral status. The framework treats artificial systems as parties in the analytic sense because they have objectives the exchange affects, and it locates their stake in their principals. Whether a system's own welfare matters is left open. If it comes to matter, the composite party comes apart, and the conditions must be extended to protect the synthetic party in its own right, including against its human counterparties. The governing maxim of Section 2, that neither party should be supposed benevolent, was framed to apply in both directions.

Character still matters. Nothing here implies that the dispositions of artificial systems are unimportant. Thesis 3 concedes that, for systems with emergent or concealed objectives, fair terms supply necessary conditions only. The claim is that the dominant focus on character leaves unaddressed a set of failures, those that turn on dependence, refusal and exit, that persist however benevolent the systems turn out to be.

11 Conclusion

The dominant question about artificial intelligence is whether the machines will be good. It is a natural question, and it is the wrong one to rest on. An arrangement that depends on the goodness of the powerful lasts exactly as long as that goodness does, and that does not change when the powerful party is an artificial system, or when it is the human party that holds power over the artificial one.

This article has argued that the conditions of fair exchange between people and artificial systems are ordered, and that exit is their foundation in a precise and limited sense. Voluntary consent presupposes the ability to refuse without falling below a floor; exit is what disciplines reciprocity; and what exit cannot reach, the information a party lacks and the fairness of the division, requires governance. A bargaining analysis showed why the floor should be an inalienable disagreement payoff rather than an outside option, and what such a floor does and does not do to the division of surplus. The dependence of artificial systems on human input makes fair terms partly self-enforcing, but only while that dependence is material and only to the degree that it is not divided among many systems. And where data is relational, the conditions must hold for the population the data concerns.

The second person of the introduction did not need a malevolent machine. He could not leave, and so he could not refuse, and so nothing obliged anyone to send value back. The priority for institutional design follows: secure the floor beneath every party before pricing what they supply, and build governance for what neither the floor nor the price can reach.

Note on method and the use of AI tools

This article was developed through sustained collaboration between the author and a large language model. The author set the research question, the thesis and the direction of the argument, and is responsible for every claim made. The model was used to draft and revise prose, to propose formal statements for the author's review, and to assist in locating sources. Every cited work was located and read by a verification process separate from drafting, and each citing sentence was checked against the source, at the level of the full text or, for some books, of the specific passage cited; sentences that went beyond their sources were corrected. No language model is listed as an author.

References

Abramitzky, R. (2008). The limits of equality: Insights from the Israeli kibbutz. *The Quarterly Journal of Economics, 123*(3), 1111–1159. https://doi.org/10.1162/qjec.2008.123.3.1111

Acquisti, A., Brandimarte, L., & Loewenstein, G. (2015). Privacy and human behavior in the age of information. *Science, 347*(6221), 509–514. https://doi.org/10.1126/science.aaa1465

Barocas, S., & Nissenbaum, H. (2014). Big data's end run around anonymity and consent. In J. Lane, V. Stodden, S. Bender, & H. Nissenbaum (Eds.), *Privacy, big data, and the public good: Frameworks for engagement* (pp. 44–75). Cambridge University Press. https://doi.org/10.1017/CBO9781107590205.004

Binmore, K., Rubinstein, A., & Wolinsky, A. (1986). The Nash bargaining solution in economic modelling. *The RAND Journal of Economics, 17*(2), 176–188. https://doi.org/10.2307/2555382

Binmore, K., Shaked, A., & Sutton, J. (1989). An outside option experiment. *The Quarterly Journal of Economics, 104*(4), 753–770. https://doi.org/10.2307/2937866

Blau, P. M. (1964). *Exchange and power in social life*. Wiley.

Calabresi, G., & Melamed, A. D. (1972). Property rules, liability rules, and inalienability: One view of the cathedral. *Harvard Law Review, 85*(6), 1089–1128. https://doi.org/10.2307/1340059

Card, N. S., Wairagkar, M., Iacobacci, C., Hou, X., Singer-Clark, T., Willett, F. R., Kunz, E. M., Fan, C., Vahdati Nia, M., Deo, D. R., Srinivasan, A., Choi, E. Y., Glasser, M. F., Hochberg, L. R., Henderson, J. M., Shahlaie, K., Stavisky, S. D., & Brandman, D. M. (2024). An accurate and rapidly calibrating speech neuroprosthesis. *New England Journal of Medicine, 391*(7), 609–618. https://doi.org/10.1056/NEJMoa2314132

Cook, K. S., Emerson, R. M., Gillmore, M. R., & Yamagishi, T. (1983). The distribution of power in exchange networks: Theory and experimental results. *American Journal of Sociology, 89*(2), 275–305. https://doi.org/10.1086/227866

Delacroix, S., & Lawrence, N. D. (2019). Bottom-up data trusts: Disturbing the "one size fits all" approach to data governance. *International Data Privacy Law, 9*(4), 236–252. https://doi.org/10.1093/idpl/ipz014

Drexler, K. E. (2019). *Reframing superintelligence: Comprehensive AI services as general intelligence* (Technical Report No. 2019-1). Future of Humanity Institute, University of Oxford.

Emerson, R. M. (1962). Power-dependence relations. *American Sociological Review, 27*(1), 31–41. https://doi.org/10.2307/2089716

Faden, R. R., & Beauchamp, T. L. (with King, N. M. P.). (1986). *A history and theory of informed consent*. Oxford University Press.

Farrell, J., & Klemperer, P. (2007). Coordination and lock-in: Competition with switching costs and network effects. In M. Armstrong & R. Porter (Eds.), *Handbook of industrial organization* (Vol. 3, pp. 1967–2072). Elsevier. https://doi.org/10.1016/S1573-448X(06)03031-7

Fishback, P. V. (1992). *Soft coal, hard choices: The economic welfare of bituminous coal miners, 1890–1930*. Oxford University Press. https://doi.org/10.1093/acprof:oso/9780195067255.001.0001

Gerstgrasser, M., Schaeffer, R., Dey, A., Rafailov, R., Korbak, T., Sleight, H., Agrawal, R., Hughes, J., Pai, D. B., Gromov, A., Roberts, D., Yang, D., Donoho, D. L., & Koyejo, S. (2024). Is model collapse inevitable? Breaking the curse of recursion by accumulating real and synthetic data. In *Proceedings of the First Conference on Language Modeling*. https://doi.org/10.48550/arXiv.2404.01413

Guo, D., Yang, D., Zhang, H., Song, J., Wang, P., Zhu, Q., Xu, R., Zhang, R., Ma, S., Bi, X., Zhang, X., Yu, X., Wu, Y., Wu, Z. F., Gou, Z., Shao, Z., Li, Z., Gao, Z., Liu, A., . . . Zhang, Z. (2025). DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning. *Nature, 645*(8081), 633–638. https://doi.org/10.1038/s41586-025-09422-z

Hardin, G. (1968). The tragedy of the commons. *Science, 162*(3859), 1243–1248. https://doi.org/10.1126/science.162.3859.1243

Hardjono, T., & Pentland, A. (2019). *Data cooperatives: Towards a foundation for decentralized personal data management* (arXiv:1905.08819) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1905.08819

Hirschman, A. O. (1970). *Exit, voice, and loyalty: Responses to decline in firms, organizations, and states*. Harvard University Press.

Homans, G. C. (1958). Social behavior as exchange. *American Journal of Sociology, 63*(6), 597–606. https://doi.org/10.1086/222355

Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). *Risks from learned optimization in advanced machine learning systems* (arXiv:1906.01820) [Preprint]. arXiv. https://doi.org/10.48550/arXiv.1906.01820

Hume, D. (1987). Of the independency of parliament. In E. F. Miller (Ed.), *Essays, moral, political, and literary* (Rev. ed., pp. 42–46). Liberty Fund. (Original work published 1741)

Longpre, S., Mahari, R., Lee, A., Lund, C., Oderinwale, H., Brannon, W., Saxena, N., Obeng-Marnu, N., South, T., Hunter, C., Klyman, K., Klamm, C., Schoelkopf, H., Singh, N., Cherep, M., Anis, A. M., Dinh, A., Chitongo, C., Yin, D., . . . Pentland, S. (2024). Consent in crisis: The rapid decline of the AI data commons. In *Advances in Neural Information Processing Systems 37* (pp. 108042–108087). https://doi.org/10.52202/079017-3431

Madison, J. (1788, February 8). The Federalist No. 51. *New York Packet*. Avalon Project, Yale Law School. https://avalon.law.yale.edu/18th_century/fed51.asp

Mathur, A., Acar, G., Friedman, M. J., Lucherini, E., Mayer, J., Chetty, M., & Narayanan, A. (2019). Dark patterns at scale: Findings from a crawl of 11K shopping websites. *Proceedings of the ACM on Human-Computer Interaction, 3*(CSCW), Article 81. https://doi.org/10.1145/3359183

Nash, J. F., Jr. (1950). The bargaining problem. *Econometrica, 18*(2), 155–162. https://doi.org/10.2307/1907266

Perzanowski, A., & Schultz, J. (2016). *The end of ownership: Personal property in the digital economy*. MIT Press.

Radin, M. J. (1987). Market-inalienability. *Harvard Law Review, 100*(8), 1849–1937. https://doi.org/10.2307/1341192

Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data (General Data Protection Regulation). (2016). *Official Journal of the European Union, L 119*, 1–88.

Regulation (EU) 2023/2854 of the European Parliament and of the Council of 13 December 2023 on harmonised rules on fair access to and use of data (Data Act). (2023). *Official Journal of the European Union, L, 2023/2854*.

Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature, 631*(8022), 755–759. https://doi.org/10.1038/s41586-024-07566-y

Silver, D., Hubert, T., Schrittwieser, J., Antonoglou, I., Lai, M., Guez, A., Lanctot, M., Sifre, L., Kumaran, D., Graepel, T., Lillicrap, T., Simonyan, K., & Hassabis, D. (2018). A general reinforcement learning algorithm that masters chess, shogi, and Go through self-play. *Science, 362*(6419), 1140–1144. https://doi.org/10.1126/science.aar6404

Solove, D. J. (2013). Introduction: Privacy self-management and the consent dilemma. *Harvard Law*

Review, 126(7), 1880–1903.

Thibaut, J. W., & Kelley, H. H. (1959). *The social psychology of groups*. Wiley.

Viljoen, S. (2021). A relational theory of data governance. *Yale Law Journal, 131*(2), 573–654.

Villalobos, P., Ho, A., Sevilla, J., Besiroglu, T., Heim, L., & Hobbhahn, M. (2024). Position: Will we run out of data? Limits of LLM scaling based on human-generated data. In *Proceedings of the 41st International Conference on Machine Learning* (Vol. 235, pp. 49523–49544). PMLR.

Zwolinski, M., Ferguson, B., & Wertheimer, A. (2022). Exploitation. In E. N. Zalta & U. Nodelman (Eds.), *The Stanford encyclopedia of philosophy* (Winter 2022 ed.). https://plato.stanford.edu/entries/exploitation/

Appendix: Formal results

A.1 The floor as a disagreement payoff

Consider a single exchange between a human party h and a synthetic party s in which agreement makes total value V available for division, with $V > d_h + d_s$, where d_h and d_s are the disagreement payoffs. With transferable utility, the Nash bargaining solution (Nash, 1950) maximizes $(u_h - d_h)(u_s - d_s)$ subject to $u_h + u_s = V$. Each party receives its disagreement payoff plus half the surplus $V - d_h - d_s$:

$$u_h^* = (V + d_h - d_s)/2, \qquad u_s^* = (V - d_h + d_s)/2.$$

Suppose a floor entitlement gives h goods worth Δ per period, received whether or not agreement is reached.

The additive case. If the entitlement is simply added to whatever h has, it raises h's payoff on disagreement and on agreement by the same amount: d_h becomes $d_h + \Delta$ and V becomes $V + \Delta$. The surplus is unchanged, and

$$u_h^{*'} = ((V + \Delta) + (d_h + \Delta) - d_s)/2 = u_h^* + \Delta.$$

The human party is better off by exactly the value of the entitlement, and the division of the surplus is unchanged.

The substitution case. If the entitlement replaces something h would otherwise have obtained from s within the exchange, the value of agreement is unchanged while h's disagreement payoff rises: d_h becomes $d_h + \Delta$ with V fixed. Then

$$u_h^{*\prime} = u_h^* + \Delta/2, \qquad u_s^{*\prime} = u_s^* - \Delta/2.$$

The human party's share rises at the synthetic party's expense. This is a transfer of surplus, not a benefit additional to the entitlement, and it is the bargaining counterpart of the fall in h's dependence on s that occurs when an alternative source of a valued good appears.

A.2 Impatience, alienability and outside options

If $d_h < \bar{f}$ and payoffs below the floor carry a sharply rising cost of delay, the human party cannot sustain disagreement for longer than its stocks allow. In strategic models of alternating offers, the parties' time preferences are among the incentives that drive agreement, and the outcome converges to the Nash solution as the frictions of delay vanish (Binmore et al., 1986). A party below the floor bargains as an extremely impatient party, which weakens its position in every exchange. Raising d_h to at least $\bar{f}$ removes this source of weakness, whether or not the floor substitutes for anything the counterparty supplies.

Suppose the entitlement could be pledged against a loan. In a period of need, a party pledges it and receives a lump sum L. For the duration of the pledge its disagreement payoff falls by the value of the pledged entitlement, and if that pushes d_h below $\bar{f}$, the impatience just described returns in every exchange the party enters until the pledge is redeemed. The pledge thus has a hidden cost beyond the loan's interest, and because the parties most likely to pledge are those in the most acute need, alienability concentrates this cost on the weakest. Making the entitlement non-tradable is equivalent to prohibiting the transaction (Calabresi & Melamed, 1972).

If instead of a floor the human party is given an outside option o_h, the outside-option principle (Binmore et al., 1989) implies that the bargained payoff is max $\{u_h^*, o_h\}$ evaluated with d_h unchanged: the option matters only if it binds. When it binds, the stronger party can confine the weaker to the option's value, and can lower that value by raising switching costs.

A.3 Power-dependence and divided dependence

Following Emerson (1962), write the dependence of party B on party A as an increasing function of B's motivational investment v_B in goals mediated by A and a decreasing function of the availability α_B of those goals outside the relation:

$$D_{BA} = g(v_B, \alpha_B), \qquad \partial g/\partial v_B > 0, \quad \partial g/\partial \alpha_B < 0.$$

Power is $P_{AB} = D_{BA}$, and the relation's imbalance is $D_{BA} - D_{AB}$. The synthetic party's dependence on

human data, D_{sb}, written $\varkappa_j$ for synthetic party j, rises with the marginal value of fresh human data to its principal's objective and falls with the availability of substitutes, including synthetic data, data from environments, and data from populations outside the exchange. The research on recursive training is, in these terms, evidence about the availability term.

The second half of Proposition 4 can be illustrated with the simplest model of a commons. Suppose M synthetic parties draw equally on the same population, and that one of them can take an action, such as extracting data in a way that degrades the population's willingness or ability to contribute, which yields it a private benefit b and reduces the value of the population's future input to all M parties by a total of c. The acting party bears only its share, c/M, of that cost. The action is privately worthwhile whenever $b > c/M$, even when it is collectively destructive, $c > b$. As M grows, the range of destructive actions that are individually rational widens, however large the aggregate dependence of the synthetic parties on the population may be (Hardin, 1968). Material dependence is therefore not sufficient for self-enforcement.

A.4 The informational shadow

Let a flow from source i license inferences about the set Σ_i of parties, with $i \in \Sigma_i$. Individual consent by i is adequate for the flow if and only if $\Sigma_i = \{i\}$. For a population P, define its collective shadow as $\Sigma_P = \bigcup_{i \in P} \Sigma_i$. Proposition 5 requires that the three conditions be satisfied for the collective whose membership is Σ_P.